A SOURCE INITIALIZATION

Immutable source memory B
fixed threshold τdet

Trusted normal set
construction / anchors / calibration

Probe and anchor bank
paired probes + source anchors

B PROSPECTIVE ADMISSION

Incoming image x_t
predict before update

Candidate proposal
top-k patches relative to B

Current adaptive memory S_t
bounded FIFO state

accept: commit $S'_t \rightarrow S_t$

B and τdet

Prospective gate
utility + probe harm
retention + margin floor

Shadow memory S'_t
candidate + FIFO eviction
exact post-eviction state

reject: keep S_t unchanged

Witness buffer W_t
candidate-disjoint; unlabeled

C DECISION-SAFEGUARDED SCORING

same incoming image x_t is scored by both channels

Counterfactual channel
source score $sB(x_t)$
uses immutable B

Adaptive channel
memory score $sM(x_t)$
uses B ∪ S_t

Decision safeguard
residual + threshold projection

Final score
source decision
preserved

data / state flow

audit / admission control